

SIH Nationals: Tech Stack and Components

AI-Powered Autonomous Biomedical Waste Segregation Bin

1 Proposed System Design

The design uses a dished tilting plate with a sweeper flap and a stopper flap mounted on an arm and hub above it. Figure 1 shows the cross-section of the bin, and Figure 2 shows the top view.

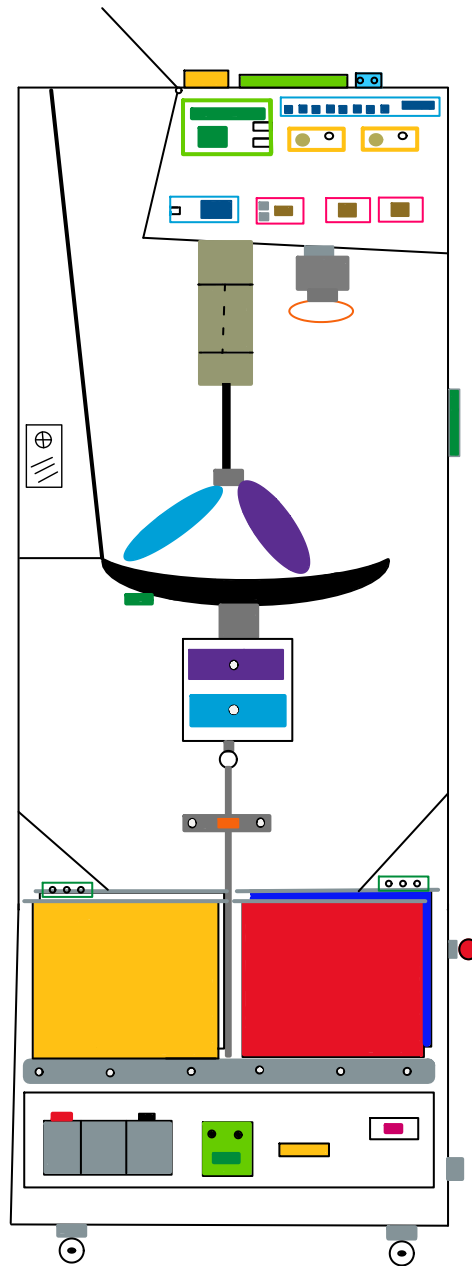


Figure 1: Medical waste segregation bin: cross-section (dished plate with sweeper and stopper flaps).

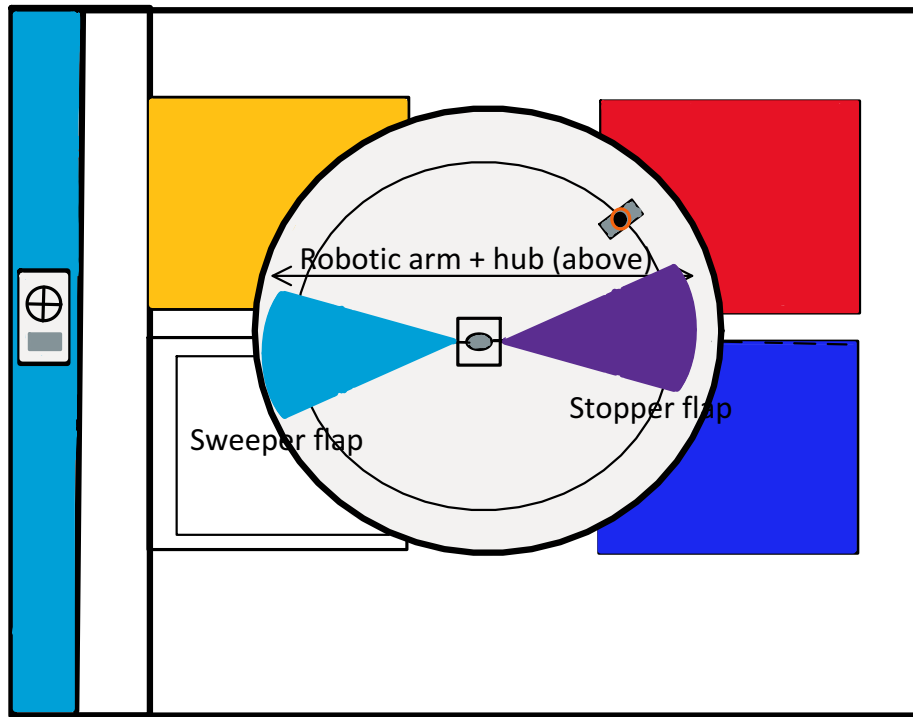


Figure 2: Top view: robotic arm and hub (above) with the sweeper flap and stopper flap over the dish.

2 Tech Stack Overview

Tech Stack Element	Primary Purpose	Role in Your Project
OpenCV	Computer Vision & Image Processing	Preprocesses waste images captured by the camera (resizing, contrast adjustment, contour detection) before classification.
NumPy	Numerical Computing & Array Handling	Handles fast matrix and pixel array manipulations coming from image frames or sensor arrays.
Inference Runtime	Lightweight Model Execution Engine	Executes your trained AI model on edge hardware efficiently (e.g., TensorFlow Lite Runtime or ONNX Runtime).
ESP32-S3 / OpenMV	Microcontroller / Edge Vision Board	The physical brain of the mobile robot; captures images, runs local inference, and controls hardware actuators.
TensorFlow	Machine Learning Framework	Used to train and export deep learning models (e.g., CNNs) for multi-class biomedical waste classification.
PySerial	Serial Communication Library	Enables data transmission between your computer/vision processor and the microcontroller (or between microcontrollers).

Tech Stack Element	Primary Purpose	Role in Your Project
Roboflow	Dataset Management & Model Pipeline	Manages your biomedical waste image dataset, applies data augmentation, annotates classes, and exports models directly.
Jupyter Notebook	Interactive Development Environment	Used for prototyping Python code, visualizing dataset statistics, training logs, and testing computer vision pipelines.
Sensors (like IR)	Object Detection & Proximity Feedback	Detects when waste is dropped into the intake chute or helps the mobile vehicle avoid obstacles.
Servo / Actuators	Mechanical Movement Control	Drives internal sorting gates, flaps, or bin-routing mechanisms to drop waste into correct compartments.

3 How to Integrate Everything into Your Project Workflow

1: Data Collection & Model Training.

1. **Roboflow:** Collect images of different biomedical waste types (e.g., syringes, cotton swabs, vials, biohazard plastics). Upload them to Roboflow to annotate, augment, and split your dataset into training, validation, and test sets.
2. **Jupyter Notebook & TensorFlow:** Train a lightweight image classification or object detection model (like MobileNetV2 or YOLO variants optimized for edge devices).
 - **Write and prototype your training scripts inside Jupyter Notebooks.**
3. **Inference Runtime:** Convert your trained TensorFlow/Jupyter model into a lighter format (like TFLite) so it can fit onto edge hardware with **minimal latency**.

2: Edge Deployment & Computer Vision (The Hardware Logic)

1. **ESP32-S3 / OpenMV:** Deploy your optimized model onto the camera-equipped microcontroller.
 - If using **OpenMV**, it has built-in Python support for basic color/shape tracking and running TensorFlow Lite models natively.
 - If using **ESP32-S3** (with a companion camera module like the **OV2640**), it handles wireless tracking, sensor inputs, and neural network inference at the edge.
2. **OpenCV & NumPy:** On a slightly larger companion processor (if needed, like a Raspberry Pi or powerful vision board), OpenCV processes incoming video frames to **isolate the waste** object via **background subtraction** or bounding box extraction before passing it to the model.

3: Hardware Control & Sorting Mechanics

1. **Sensors (IR/Proximity):** Position an IR sensor near the waste entry chute. When a piece of waste triggers the sensor, it signals the ESP32-S3 to snap a photo.
2. **PySerial:** If your computer or primary vision board processes the classification result separately from the motor driver, PySerial passes sorting commands (e.g., `SORT_INFECTIOUS_RED` or `SORT_SHARPS_YELLOW`) over serial communication.
3. **Servo Motors:** Based on the command received, the microcontroller actuates specific servo motors or **stepper gates** to tilt the collection tray or open the correct internal compartment.

4: Mechanical & Product Design Integration

- **Autodesk Fusion:** Bring all your electronic layouts, chassis designs, wheel housings, and internal segregation compartments together into a single unified 3D assembly. Use Fusion's simulation tools to test structural loads, and generate manufacturing-ready files for the final prototype.

4 Additional Tech Stack Items Required for Project

- **Microcontroller Firmware / IDE:** **Arduino IDE** or **PlatformIO** (used to write, compile, and flash code onto the ESP32-S3).
- **Database & Cloud Backend:** **Firebase Realtime Database** or **Node.js with PostgreSQL** (to store medical-waste collection logs, compartment weight levels, timestamps, and route data for the digital traceability requirement).
- **Dashboard / Web UI:** **React.js** or a simple HTML/Chart.js dashboard (so hospital staff can monitor bin capacities, waste category counts, and system battery levels in real time).
- **Power Management Hardware:** **BMS (Battery Management System)** module, step-down buck converters (e.g., LM2596), and Li-ion/LiFePO4 battery packs to power the motors and ESP32 safely.
- **Motor Driver:** **L298N** or **BTS7960** (to control DC drive motors for the mobile robot chassis movement).

5 Autodesk Fusion Tech Stack & Workspace Mapping

1. Autodesk Fusion Electronics (ECAD) — *Alternative to External Circuit Design Tools*

- **What it does:** Instead of designing your microcontroller breakout or sensor connection boards elsewhere, Fusion has a fully integrated **Electronics Workspace**.
- **Role in Project:** You can design your custom PCB schematic for connecting the ESP32-S3, IR sensors, and servo motors, run **SPICE simulations** to check electrical behavior, and route the PCB layout directly inside Fusion.

2. Fusion Generative Design & Autodesk AI — *Alternative to Manual Structural Trial-and-Error*

- **What it does:** Uses cloud-based AI algorithms to automatically generate lightweight, high-strength structural brackets, chassis components, or internal segregation carousels based on loads and manufacturing constraints (like 3D printing or CNC machining).
- **Role in Project:** You can use it to design a lightweight, optimized robotic chassis framework that holds heavy internal bins while minimizing battery strain.

3. Fusion CAE / Simulation Workspace — *Alternative to External Stress Analysis Software*

- **What it does:** Provides finite element analysis (FEA) and motion simulation.
- **Role in Project:** Test structural loads on the sorting mechanisms and check how the vehicle frame handles weight distribution when bins fill up with medical waste.

4. Fusion CAM Workspace — *Direct Manufacturing Handoff*

- **What it does:** Computer-Aided Manufacturing toolpaths.
- **Role in Project:** Generates G-code directly from your 3D models if you plan to CNC machine parts or 3D print the internal segregation compartments and outer casing.

6 Complete Sensors and Components Required

1. Vision & Object Identification

- **ESP32-S3 Cam / OpenMV H7 Module:** Acts as the edge-vision brain. Equipped with a camera sensor (like the **OV2640**), it captures live images of the medical waste dropped into the intake chute and runs local AI model inference (TFLite) to classify items (e.g., sharps, infectious soft waste, pharmaceuticals).



- **Infrared (IR) Obstacle / Entry Sensor:** Positioned right at the entry hopper. When a piece of medical waste is dropped in, the IR beam breaks, sending an interrupt signal to the microcontroller to wake up the camera and snap a picture for classification.

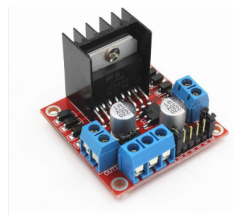


2. Navigation & Chassis Mobility

- **DC Geared Motors or BLDC Motors:** Drive the wheels of your mobile robot base.



- **Motor Driver (e.g., L298N or BTS7960):** Acts as the muscle amplifier between the low-voltage control signals of the microcontroller and the high-current requirements of the drive motors.



- **Ultrasonic Sensors (HC-SR04) / LiDAR (optional):** Mounted around the mobile robot chassis to handle real-time obstacle avoidance and wall detection as the robot navigates hospital corridors.
- **Wheel Encoders:** Track wheel rotations to provide precise odometry feedback for movement tracking and navigation mapping.



3. Segregation & Mechanical Actuation

- **High-Torque Servo Motors (e.g., MG996R) or Stepper Motors:** Control the internal sorting gates, sliding floors, or rotating carousels. Once the AI vision system identifies the waste type, the microcontroller moves the specific servo to align the internal chute with the correct biohazard compartment (e.g., Yellow bag for infectious, Red for plastic, White/Blue for sharps).

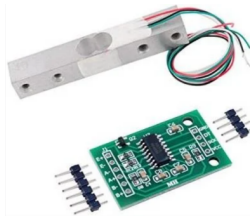


- **Inductive Proximity Sensor:** Placed near the sorting track to detect metallic elements (such as needles, scalpels, or metal surgical tools) as a reliable secondary check alongside the camera model.

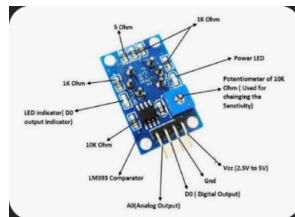


4. Traceability, Monitoring & Safety

- **Load Cells (Weight Sensors) with HX711 Amplifier:** Placed underneath each internal waste compartment to measure weight in real time, alerting hospital staff when specific bins are approaching capacity.



- **Gas / Hazard Sensor (e.g., MQ-135 or MQ-7):** Detects harmful volatile organic compounds or chemical fumes leaking from improperly sealed medical waste to maintain safety compliance.



- **OLED / 16x2 LCD Display:** Mounted externally on the robot to display real-time operational status, active compartment fill levels, and system health.



- **Buzzer / Status LEDs:** Provide audio-visual warnings if a hazard threshold is crossed or if a compartment error occurs.

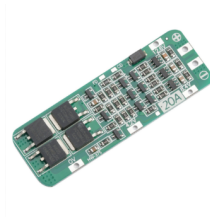


5. Power Management

- **Li-ion or LiFePO4 Battery Pack:** Provides a stable mobile power source for the entire system.



- **Battery Management System (BMS) Board:** Protects the battery from overcharging, deep discharge, and short circuits.



- **Buck Converters (e.g., LM2596):** Step down the main battery voltage (e.g., 12V) to safe regulated levels (like 5V and 3.3V) for sensitive electronics, microcontrollers, and sensors.

